

Difficult-Case Error Audit — License Plate Detection

Manual review of the 40 lowest mean-IoU test images, ordered from worst (rank 01, IoU 0.000) to least severe (rank 40, IoU 0.810). Each case was opened manually and visually classified against the four error categories below.

Summary

Annotation error	6 of 40 cases
Missed plate	4 of 40 cases
Localization error	24 of 40 cases
Correct detection	6 of 40 cases

Per-Image Classification

Filename	Category	Explanation
01_iou_0.000_3-13734.jpg	Missed plate	White Dodge Challenger's rear plate ('3 13734 SHARJAH') is clearly visible and ground-truthed, but the model produced no detection anywhere in the frame — zero overlap.
02_iou_0.443_9-20202.jpg	Annotation error	Source photo has a panorama/motion-blur stitching glitch that duplicates the Mercedes' plate region. Ground truth labels the glitched duplicate as well as the real plate; only the real plate is matched by a prediction.
03_iou_0.455_I-55529.jpg	Annotation error	A blue highway exit sign fragment ('36') in the background is labelled as a ground-truth plate. It is not a plate, so no prediction matches it, even though the Audi R8's real plate is detected accurately.
04_iou_0.462_48981.jpg	Blank plate	A second plate on the vehicle is blurred and not detected by the model.
05_iou_0.470_T-2317.jpg	Annotation error	Same recurring blue road-sign fragment ('34') mislabeled as a ground-truth plate; the Tesla's actual plate is detected correctly.
06_iou_0.470_34773.jpg	Blank plate	Small rear blurred plate of the white SUV in the background is ground-truthed but not picked up by the detector, while the foreground sedan's plate is detected correctly.
07_iou_0.473_L-64950.jpg	Annotation error	Blue road-sign fragment ('36') again mislabeled as a ground-truth plate; the Abarth's real plate detection is accurate.
08_iou_0.482_R-97898.jpg	Annotation error	Same recurring road-sign mislabeling ('36') pulls the mean down, while the Bentley's actual plate is well matched.
09_iou_0.602_18562166.jpg	Annotation error	Road sign fragment ('34') mislabeled as ground truth again; both real plates in the scene (Infiniti and Maserati) are detected well.
10_iou_0.628_WhatsApp...3334.jpg	Correct detection	Ground-truth box is drawn noticeably larger than the actual plate (includes surrounding bumper trim), while RF-DETR's tighter box sits correctly on the plate itself; the loose annotation, not the model, causes the low IoU.

Filename	Category	Explanation
11_iou_0.697_22317.jpg	Correct detection	Ground-truth box is drawn noticeably larger than the actual plate (includes surrounding bumper trim), while RF-DETR's tighter box sits correctly on the plate itself; the loose annotation, not the model, causes the low IoU.
12_iou_0.703_L-8453.jpg	Localization error	Predicted box boundary is slightly offset from the tightly-drawn ground-truth box on the Range Rover's plate.
13_iou_0.707_1-76091.jpg	Correct detection	Ground-truth box extends above the plate into the chrome bumper trim, while the prediction is tightly and correctly fit to just the plate.
14_iou_0.715_5161.jpg	Localization error	Predicted box is shifted right, cutting off the left edge of the 'CLASSIC' plate relative to the ground-truth box.
15_iou_0.717_Q-36930.jpg	Localization error	Predicted box is marginally narrower than the ground-truth box along the bottom edge of the Aston Martin's plate.
16_iou_0.719_80900.jpg	Correct detection	The Explorer's plate is blurred and the model's box; a second, clearly visible taxi plate nearby has no ground-truth box in this crop.
17_iou_0.736_Q-59744.jpg	Localization error	Predicted box is slightly smaller than the ground-truth box on the BMW's plate.
18_iou_0.739_U-55077.jpg	Correct detection	Ground-truth box is much bigger than the plate, while the predicted box is more accurate.
19_iou_0.741_W-96030.jpg	Correct detection	Ground-truth box includes the dealer-sticker text below the plate ('Al Nabooda Automobiles'), while the prediction is correctly confined to just the plate itself.
20_iou_0.750_Q-70077.jpg	Localization error	Ground-truth box is much bigger than the plate, while the predicted box is more accurate.
21_iou_0.752_Q-55254.jpg	Localization error	Minor boundary mismatch on the Alfa Romeo Giulia's front plate.
22_iou_0.754_Q-9563.jpg	Localization error	Minor boundary mismatch on the AMG GT's plate.
23_iou_0.757_D-53980.jpg	Localization error	Minor boundary mismatch on the Bentley Continental's plate.
24_iou_0.762_Q-14771.jpg	Localization error	Minor boundary mismatch on the Aston Martin DB9's plate.
25_iou_0.768_Q-70585.jpg	Localization error	Minor boundary mismatch on the Toyota Highlander's plate.
26_iou_0.783_WhatsApp...53617.jpg	Correct detection	Ground-truth box is substantially larger than the plate itself (includes reflective bumper chrome around it), while the model's prediction is tightly and correctly fit to the plate boundary.
27_iou_0.784_Q-71423.jpg	Localization error	Minor boundary mismatch on the Land Cruiser's plate.
28_iou_0.787_W-81473.jpg	Correct detection	Correct prediction of the Rolls-Royce Wraith's plate.

Filename	Category	Explanation
29_iou_0.787_L-24331.jpg	Missed plate	A second, partially cropped plate from a neighboring vehicle at the image's left edge is visible and detected by the model but is not included in the ground-truth annotations.
30_iou_0.788_Q-1213.jpg	Localization error	Minor boundary mismatch on the Aston Martin DBS's plate.
31_iou_0.789_Q-60323.jpg	Localization error	Minor boundary mismatch on the Peugeot RCZ's plate.
32_iou_0.790_Q-12782.jpg	Localization error	Minor boundary mismatch on the Porsche 911's plate.
33_iou_0.791_6-73186.jpg	Correct detection	Ground-truth box is slightly larger than the plate (includes surrounding bumper trim/bracket), while the prediction fits the plate itself more tightly.
34_iou_0.798_P-10848.jpg	Localization error	Minor boundary mismatch on the GMC Sierra's plate.
35_iou_0.799_10-49387.jpg	Localization error	Minor boundary mismatch on the small, distant Altima plate.
36_iou_0.804_Q-65301.jpg	Localization error	Minor boundary mismatch on the Mustang's plate.
37_iou_0.806_3-38321.jpg	Localization error	Minor boundary mismatch on the Ford Taurus's plate.
38_iou_0.807_T-14396.jpg	Localization error	Minor boundary mismatch on the Cadillac Eldorado's plate.
39_iou_0.807_WhatsApp...82170.jpg	Correct detection	Ground-truth box is drawn much larger than the plate itself, with wide margins on all sides, while the model's prediction correctly and tightly outlines just the plate.
40_iou_0.810_B-10074.jpg	Localization error	Prediction and ground-truth boxes are both close to the plate but not pixel-identical; small edge offset accounts for the residual gap from a perfect IoU.